# Business Finance  Sept 4th

Capital = money

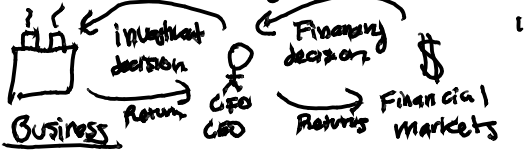
How?

CFO
- investment decision
- finance decision

## Finance is the study

allocation of scarce resources

- sub field of Econ
- what I can buy with capital




Make connections you didn't know

Help the class

How to get capital to where

first month
↓
Understand Risk and how we price it.
then
↓
investment decision
↓
financing decision

first 4 weeks  doc, rules of returns risk

fundamental equation of all finance

4 equations

tables

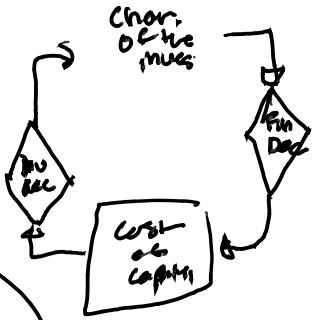
- Forecasts
- Dallin McLean
- Tutoring program

Discounting
Risk
Next NPV

CFO

Assets
equity liability
makes depreciation

- Balance sheet
- income statements
- cash flow




model oriented system

on time

5% sets
25% cases
20 participation
20 mid
30 final

6 problem sets
6 cases
2 exams

open analysis

models only good as assumptions
defend (1)
decide more correctly (2)
make good analysis (3)

Testing center

100% on the exam

After class do an outline

1 one key thing

how do they fit

clean problem sets

Discount rate = conversion rate = apple

Discount rate = the rate of return investors could earn if they invested in the next best project over a similar time & with similar risk

↳ opportunity cost

$$PV = \sum_{t=0}^{\infty} E(CF_t)$$

$$FV_t = (PV_0)(1+r)^t$$

$$\frac{(FV)_t}{(1+r)^t}$$

Constant perpetuity

$$PV = \frac{CF}{r}$$

growing perpetuity

$$PV = \frac{CF_t + 1gr}{r - g}$$

$$\frac{\text{cash flow next yr}}{r - g}$$

Cash flow

R

Pricing a bond

$r_i = r_f + \text{risk}$

inflation

Bond - 3 years left

- coupon → interest rate 5%

YTM 7%

$1000 \cdot (.05) = 50 = 1050 = 105$

↑ interest rates ↓ financial instruments

$$\frac{50}{1+.07} + \frac{50}{(1.07)^2} + \frac{50}{(1.07)^3} = \quad T=3$$

$$PV = \frac{50}{.07}$$

$\infty$

$r_i = r_f + \text{risk}$ → sigma squared $\sigma^2$

→ match cash flow time

if ∞ $\underline{10}$ or $\geq 30$

≠ low prob of success

= Variance or uncertainty

we dont like risk

we like returns

↑ not risky is just bad

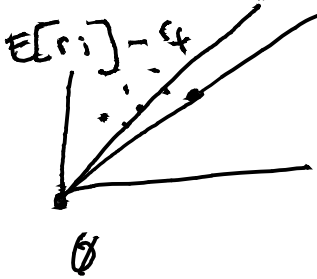
3000 = certain

$$\text{Sharpe Ratio} = \frac{\text{expected return for any investment} - r_f}{\sigma_i}$$

$\sigma_i$ = measures risk

$\sigma_i$ = standard deviation

Discount rate = opportunity cost




capital allocation line

CAL Line

slope = sharp ratio $\frac{\text{rise}}{\text{run}}$ $\frac{E[r_i] - r_f}{\sigma_i}$

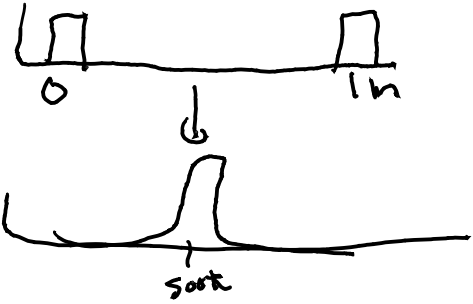
$\sigma_i$

steeper lines

higher sharp ratios

4% no excess return

diversification = free lunch




diversification makes risk go down t
and returns being guaranteed

Diversification

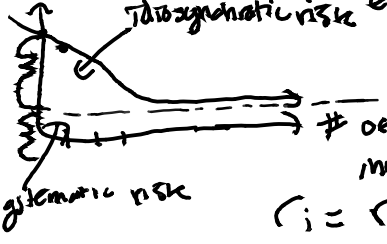
investors don't demand return when diversified

$\boxed{\text{Diversifiable risk is not priced}}$

Belongs to each security

diversifiable risk

idiosyncratic risk




systematic risk

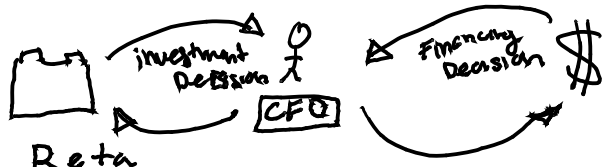
systematic risk

$r_i = r_f + \text{risk}$

p

Cases

1. Make some assumptions and why?
2. Did you do the analysis correctly
3. Did you draw reasonable conclusions




Beta

$\beta_{Debt}$, $\beta_{Equity}$

$\beta$ is the covariance

SP500 against $\beta$ of my equity

$$\boxed{r_i = r_f + \beta_i (r_m - r_f)} \leftarrow \text{CAPM}$$

Capital Asset Pricing Model

Discount Rate

Accounts for time & risk

$r_f + \beta_i (r_m - r_f)$

$r_f$ → Time → Investment inflation

$\beta_i$ → Systematic Risk

$(r_m - r_f)$ → + Risk

$\beta_i (r_m - r_f)$

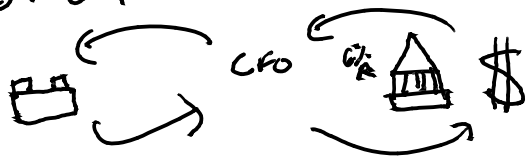
$\beta_i$ → quantity of risk

$(r_m - r_f)$ → price of holding one unit of risk

$\beta_i (r_m - r_f)$ → spread

"Spread" above the risk free rate

3 year 6% coupon




Sells for 96.23 = coupon rate is not competitive = not par

$R_D$ = Discount Rate & yield to maturity = 6% = too low

96.23 = PV

what's $r_D$?

PV = x periods PV PMT

= RATE(3, 960, 60, 0)

| 0 | 1 | 2 | 3 |
|---|---|---|---|
| 962.30 | 60 | 60 | 1060 |

$$962.30 = \frac{60}{1+x} + \frac{60}{(1+x)^2} + \frac{1060}{(1+x)^3}$$

= RATE(periods, PV, PMT, FV, 0)

$$\boxed{r_D = 7.45\%}$$

$r_f = 4.8\%$
$MRP = 5\%$
$(r_m - r_f) = 5\%$

$$r_i = r_f + \beta_i (r_m - r_f)$$

$$\begin{array}{r} 7.45 \\ -4.60 \\ \hline 2.65 \end{array}$$

$$r_i = 4.8\% + x(5\%)$$

$$y = 4.8 + x(5\%)$$

$$\frac{7.45}{5} - \frac{4.6}{5}$$

$$\frac{7.45\%}{5\%} \quad \frac{4.8\%}{5\%} + \frac{x(\cancel{5\%})}{\cancel{5\%}}$$

$x$

$r_i = r_f + \beta_i (r_m - r_f$ Debt is less risky than equity

$$\boxed{\beta_i = \frac{r_i - r_f}{r_m - r_f}}$$

Systematic Risk

interest rate risk
Default risk

Bond PMT

$r_f = 2.5\%$

$$\boxed{r_D = 2.5 + .53(5)}$$

$r_D = 5.15\%$ ↑ par

$$P = \frac{60}{(1.0515)} + \frac{60}{(1.0515)^2} + \frac{1060}{(1.0515)^3}$$

$P = 102.3$

$P = 102.3.10$

or 102.31

Dividends = \$2.36/sh

What price are you willing to pay for $V_0$?

$P_1 = 360$

$\beta_v = .92$

2 yr treasury = 3.57%

$r_f = 3.57\%$

MRP = 5%

$x = 3.57 + .92(5)$

$$P_0 = \frac{2.36}{1.0817} + \frac{352.36}{(1.0817)^2}$$

$$\text{spread} = r_D - r_f$$

$$r_D = r_f + \beta_D (r_m - r_f)$$

$\downarrow$ treasury $\uparrow$ spread $\Rightarrow$ Fear $\uparrow$

sum of cash flows

YTM

$$107.7 = \sum_{t=0}^{8} \frac{CF}{(1+r)^t}$$

coupon rate 7.75

$$\frac{1}{7750} \cdots \frac{8}{1077.50}$$

IRR = cash flows $-100, t_1, t_2$

Rate

Yield

YTM